MANUAL TESTING GUIDE

NSPMS User Guide

National Startup Performance Monitoring System · 4 Personas · All 16 Features

🟢 FOUNDER — HEALTHY

founder1@nspms.in

NovaSaaS Technologies · SaaS · Series A

password: password123

🔴 FOUNDER — AT-RISK

founder2@nspms.in

MedQuick Health · Healthtech · Seed

password: password123

💼 INVESTOR

investor@nspms.in

Full Investor Dashboard · Compare startups

password: password123

🛡️ ADMIN

admin@nspms.in

Admin Dashboard · Manage all users

password: password123

NSPMS — Manual Testing User Guide

4 Personas · Every Feature Covered

*Open the app at **http://localhost:5173** · Backend runs at **http://localhost:5000***

🔑 Credentials at a Glance

#	PERSONA	EMAIL	PASSWORD	ROLE	STARTUP	PLAN
1	**Arjun Mehta**	`founder1@nspms.in`	`password123`	Founder	NovaSaaS Technologies (SaaS / Series A)	Enterprise
2	**Priya Nair**	`founder2@nspms.in`	`password123`	Founder	MedQuick Health (Healthtech / Seed)	Free
3	**Rohit Sharma**	`investor@nspms.in`	`password123`	Investor	—	Enterprise
4	**Sunita Das**	`admin@nspms.in`	`password123`	Admin	—	Enterprise

Feature Coverage Matrix

FEATURE	PERSONA 1 (ARJUN)	PERSONA 2 (PRIYA)	PERSONA 3 (ROHIT)	PERSONA 4 (SUNITA)
Login / Logout	✅	✅	✅	✅
Register (signup)	—	✅	—	—
Account Lock (5 wrong passwords)	✅	—	—	—
Startup Profile	✅	✅	—	—
Dashboard (healthy)	✅	—	—	—
Dashboard (at-risk + alerts)	—	✅	—	—
Submit New Metrics	✅	✅	—	—
Metrics History & Audit	✅	✅	—	—
Vitality Score / Health Score	✅	✅	—	—
Smart Alerts (dismiss)	✅	✅ 🔴 Critical	—	—
AI Forecasting	✅	—	—	—
AI Risk + Benchmarking	✅	—	—	—
AI Advisor (chatbot)	✅	—	—	—
Milestones (completed)	✅	—	—	—
Milestones (overdue, OKRs)	—	✅	—	—
Custom KPI Builder	✅	✅	—	—
Scenario Simulation	✅	—	—	—
Reports & CSV Export	✅	—	—	—
Plans / Subscription page	—	✅	—	—
Investor Dashboard	—	—	✅	—
Investor Compare (2 startups)	—	—	✅	—

FEATURE	PERSONA 1 (ARJUN)	PERSONA 2 (PRIYA)	PERSONA 3 (ROHIT)	PERSONA 4 (SUNITA)
Admin Dashboard	—	—	—	✅
Admin Startup Detail	—	—	—	✅
Admin Change Subscription	—	—	—	✅

🟢 Persona 1 — Arjun Mehta (Healthy Founder)

Email: `founder1@nspms.in` | **Password:** `password123`
Startup: NovaSaaS Technologies | SaaS · Series A · Bengaluru · 45 employees
Mock Data: 8 months of metrics (Aug 2024 → Mar 2025), ₹8Cr total funding

Step-by-Step Walkthrough

1. Login

1. Go to `http://localhost:5173`
2. Enter `founder1@nspms.in` / `password123` → click **Login**
3. ✅ Redirected to `/dashboard`

> ***Test Account Lock:*** *Enter wrong password 5 times → see the lockout error. Wait 15 min or run* `npm run seed` *to unlock.*

2. Dashboard (`/dashboard`)

- **Vitality Score Gauge** — should show ~82–88/100 (green)
- **Risk Badge** — **Low** 🟢
- **Funding Readiness** — >80%
- **KPI Cards:** Revenue ₹6.48L, Burn ₹0 (profitable), Runway 26 months, Users 3,350, Churn 3.1%, LTV/CAC 18.7x
- **5 Charts:** Revenue Growth (upward trend), User Growth (bar chart), Revenue vs Expenses (converging), Burn Rate Trend, Expense Distribution (pie)
- **Benchmark bar:** Compare NovaSaaS vs industry averages *(should be well above)*
- **Alerts panel:** Should show 0 critical alerts (healthy startup)

3. Submit Metrics (`/metrics/new`)

1. Click **"+ Submit Metrics"** in the nav
2. Select period → e.g., `2025-04`
3. Fill Financial: Revenue `720000`, Expenses `535000`, Burn `0`, Runway `28`, Funding `8000000`
4. Fill Operational: Active Users `3800`, New Users `780`, CAC `890`, LTV `17800`, Churn `2.9`, GMV `2160000`, Cities `14`
5. Fill Innovation: Patents Filed `5`, Granted `2`, Trademarks `4`, R&D `118000`
6. Fill Impact: Direct Jobs `48`, Women `19`, Rural `8`, Exports `650000`

7. Click **Submit**
8. ✅ Redirected to Dashboard — charts update with new month, new vitality score computed

4. Metrics History & Audit Trail (`/history`)

- Table shows all 8+ submitted periods in reverse order
- Each row: Period, Revenue, Burn, Runway, Users, Churn, CAC, LTV/CAC, Health Score
- **Audit Log tab** — see log of all actions (metrics submitted, profile saved, alerts dismissed)

5. Startup Profile (`/profile`)

1. Click profile icon or nav → **Profile**
2. See pre-filled: NovaSaaS Technologies, SaaS, Series A, Bengaluru, 45 employees
3. Edit Company Description → save → ✅ Profile updated toast
4. Edit Twitter/LinkedIn links → save

6. Milestones & OKRs (`/milestones`)

1. View **All** tab — 9 milestones shown
2. Click **Milestones Only** — 5 simple milestones, 5 with ✅ Completed badge
3. Click **OKRs Only** — 3 OKRs with Key Results and progress bars
4. **Add a new milestone:**
   - Click **+ Add Milestone**
   - Title: `Launch API Marketplace`, Category: `Product`, Deadline: `2025-06-30`, Description: any text
   - Toggle **OKR mode** → add Key Result: `API Partners`, Target `20`, Unit `partners`
   - Save → appears in the list
5. **Update progress:** drag the completion slider on any in-progress milestone
6. **Mark complete:** click ✓ **Complete** on the new milestone → timestamp shown
7. **Overall progress bar** at top updates automatically

7. Custom KPI Builder (`/custom-kpis`)

1. See 4 pre-built KPIs: LTV/CAC, Revenue per Employee, Burn Multiple, NRR
2. Each shows a line chart trending across 8 months
3. **Add a new KPI:**
   - Click **+ Add KPI**
   - Name: `GMV per City`, Formula: `gmv / citiesServed`, Unit: `₹`, Description: any
   - Save → card with empty chart appears
4. **Log a value for a period:**
   - Click **+ Log Value** on the new KPI
   - Period: `2025-03`, Value: `154615`
   - Save → chart updates

8. AI Forecasting (`/forecasting`)

1. Navigate to **Forecasting**
2. Wait for AI to compute (first visit ~5–10 seconds with OpenRouter key)
3. See 🤖 **AI-Enhanced** badge + cache timestamp

4. **4 charts:** Revenue Forecast, Expense Forecast, User Growth, Runway Projection
   - Solid lines = historical actual data (Aug 2024 → Mar 2025)
   - Dashed lines = AI predictions (Apr → Sep 2025)
5. **AI Insights panel** — 5 bullet points of AI analysis
6. **Strategic Outlook** — 2–3 sentence narrative
7. **Critical Action** — single most important 30-day action
8. **Confidence level** badge (should be `high` with 8 months of data)
9. Submit new metrics → revisit Forecasting → AI regenerates (cache invalidated)

9. AI Intelligence Center (`/ai-risk`)

Tab 1 — Risk Prediction:

1. AI Risk Score displayed (~25–35 for NovaSaaS — Low risk)
2. **Trend Analysis** paragraph
3. **Risk Predictions grid** — 6 cards with probability, timeframe, impact
4. **AI Recommended Actions** — numbered list
5. Click 🔄 **Re-analyze** button → forces fresh AI call (bypasses cache)

Tab 2 — AI Benchmarking:

1. **Competitive Position** — should show "Top Quartile" or "Above Average"
2. **Peer Comparison** summary sentence
3. **Strength Areas** — metrics where NovaSaaS leads peers (green badges with + %)
4. **Areas to Improve** — red badges with - %
5. **Competitive Edge** narrative
6. **AI-Detected Opportunities**

10. AI Advisor (`/advisor`)

1. **Auto-recommendations panel** loads on page open (5 pre-generated cards)
2. Type a custom question: `"When should we start our Series B fundraise?"`
3. Press **Send** → AI responds with data-grounded answer citing actual numbers
4. Ask follow-up: `"What is our biggest operational risk right now?"`
5. Ask: `"How do we reduce CAC from ₹920 to ₹700?"`
6. **Data Points Used** chips appear below each AI answer

11. Scenario Simulation (`/simulation`)

1. **Sliders:** Revenue Change `+30%`, Expense Reduction `10%`, Additional Funding `₹50L`, Churn Improvement `20%`
2. Click 🔬 **Run Simulation**
3. **Radar Chart** — amber shape (simulated) vs gray (current) — revenue and runway axes should expand
4. **Comparison Table** — Current vs Simulated vs Change for all 6 metrics
5. Try extreme: Revenue `-40%` → see health score drop sharply → Risk Level changes to High/Critical

12. Reports & Export (`/reports`)

1. Click **Performance Summary Report** card → report generates in-browser
2. See: company header, health score, latest KPIs, all historical metrics table, milestones, alerts

3. Click 🖨️ **Print / Save as PDF** → browser print dialog opens; save as PDF
4. Click 📄 **Export CSV** → `.csv` file downloads with all metrics history (open in Excel)

13. Logout

- Click user avatar → **Logout** → redirected to `/login`

🔴 Persona 2 — Priya Nair (At-Risk Founder)

Email: `founder2@nspms.in` | **Password:** `password123`
Startup: MedQuick Health | Healthtech · Seed · Chennai · 12 employees
Mock Data: 6 months of declining metrics, 2-month runway, 21% churn

Step-by-Step Walkthrough

1. Register as a New User (Test Signup Flow)

> *This tests the registration feature. You can either register a brand-new account OR log in directly with founder2@nspms.in.*

To test Registration:

1. On Login page, click **"Register"** or **"Sign Up"** toggle
2. Fill: Name `Test Founder`, Email `testfounder@nspms.in`, Password `password123`
3. Company Name: `TestCo`, Sector: `Fintech`, Stage: `Idea`, City: `Mumbai`
4. Click **Register** → ✅ auto-logged in, redirected to Dashboard (empty, no data yet)
5. Logout and log back in as `founder2@nspms.in` to see the pre-seeded at-risk data

2. Login as Priya

1. Email `founder2@nspms.in` / `password123` → Login
2. ✅ Dashboard loads immediately

3. Dashboard — At-Risk View (`/dashboard`)

- **Vitality Score** — ~18–28/100 (red/critical)
- **Risk Badge** — **Critical** 🔴
- **Funding Readiness** — <20%
- **KPI Cards:** Revenue ₹58K (low), Burn ₹1.87L/mo, Runway **2 months** 🚨, Users 270, Churn **21%** 🚨, LTV/CAC **0.86x** 🚨
- **Smart Alerts Panel** — should show 3–4 critical/warning alerts:
  - 🚨 Critically Low Runway (2 months)
  - 🚨 High Churn Rate (21%)
  - 🚨 Unhealthy LTV/CAC Ratio (<1x)
  - ⚠️ Revenue Decline (>20% drop)
- **Dismiss an alert:** click ✕ on one → alert disappears → refresh to confirm it's gone
- **Revenue vs Expenses chart** — expenses dwarf revenue (large gap visible)
- **Benchmark comparison** — shows MedQuick well below Healthtech/Seed industry averages

4. Milestones — Overdue View (`/milestones`)

1. All tab: 7 milestones
2. **Overdue items** highlighted in red — deadlines past but not completed:
   - "Onboard 500 Active Patients" (due Dec 2024) — 70%
   - "Integrate 10 Specialist Doctors" (due Nov 2024) — 40%
   - "Telemedicine License" (due Dec 2024) — 55%
   - "Series Seed Fundraise" (due Feb 2025) — 20%
   - "Hire Head of Medical Operations" (due Jan 2025) — 0%
3. **OKRs Only** tab: see 2 OKRs — "Reduce Churn" OKR shows current 21% vs target 10%
4. Update the "Reduce Churn" OKR key result slider — drag to a new value → saves

5. Plans Page (`/plans`)

> *Priya is on the Free plan — the Plans page shows upgrade options*

1. Navigate to `/plans` (subscription page)
2. See **current plan: Free** highlighted
3. See Pro plan features listed (Forecasting, AI Advisor, Reports)
4. See Enterprise plan features (Simulation, Custom KPIs, Benchmarking)
5. *(Plan change is done by Admin — Priya can see but Admin upgrades it)*

6. Submit Metrics (Add Another Month)

1. Go to `/metrics/new`
2. Submit `2025-04` metrics — use even more dire numbers (Runway `1`, Churn `23%`) to trigger more alerts
3. ✅ New alerts generated — dashboard re-renders with fresh critical score

7. Custom KPIs (`/custom-kpis`)

1. See 3 pre-built KPIs: LTV/CAC, Consultation Completion Rate, Doctor Utilization Rate
2. All 3 show downward trending line charts (declining health)
3. Add a new KPI: `Patient Repeat Rate`, Formula: `repeat_visits / total_visits`, Unit: `%`
4. Log value for `2025-03` : `34`

8. Metrics History (`/history`)

1. See 6 months of data showing declining revenue, rising burn, shrinking runway
2. Audit log shows all metric submission entries

💼 Persona 3 — Rohit Sharma (Investor)

Email: `investor@nspms.in` | **Password:** `password123`

Step-by-Step Walkthrough

1. Login

1. Email `investor@nspms.in` / `password123` → Login
2. ✅ Redirected to `/investor` (Investor Dashboard)
3. *Trying to visit `/dashboard` directly → redirected back (role-protected)*

2. Investor Dashboard (`/investor`)

- **Startup Cards:** See both **NovaSaaS Technologies** and **MedQuick Health**
- Each card shows: Health Score, Revenue, Active Users, Runway, Revenue Growth %, Risk Badge, Funding Readiness bar

3. Search & Filter

1. **Search box:** Type `Nova` → only NovaSaaS shows
2. Clear → Type `Med` → only MedQuick shows
3. **Filter by Sector:** Select `SaaS` → only NovaSaaS
4. **Filter by Stage:** Select `Seed` → only MedQuick
5. Clear all filters → both appear

4. Sort Startups

1. **Sort by Health Score** → NovaSaaS (82+) appears before MedQuick (22)
2. **Sort by Revenue** → NovaSaaS first (₹6.48L vs ₹58K)
3. **Sort by Funding Readiness** → NovaSaaS first

5. Side-by-Side Comparison (Key Feature!)

1. Click **NovaSaaS Technologies** card → blue ring appears (selected)
2. Click **MedQuick Health** card → second blue ring
3. Click ⚖️ **Compare Selected** button
4. **Comparison panel opens:**
   - Side-by-side: Health Score (82 vs 22), Revenue (₹6.48L vs ₹58K), Users (3350 vs 270), Runway (26mo vs 2mo), Burn (₹0 vs ₹1.87L), Churn (3.1% vs 21%), Team (45 vs 12), City (Bengaluru vs Chennai)
   - **Radar Chart:** Two overlapping spider charts (indigo = NovaSaaS, violet = MedQuick) across 5 dimensions
5. Close panel → deselect cards

🛡️ Persona 4 — Sunita Das (Admin)

Email: `admin@nspms.in` | **Password:** `password123`

Step-by-Step Walkthrough

1. Login

1. Email `admin@nspms.in` / `password123` → Login
2. ✅ Redirected to `/admin` (Admin Dashboard)

3. *Trying* `/dashboard` *or* `/investor` *→ role-protected, redirected*

2. Admin Dashboard (`/admin`)

Platform Overview Stats:

- Total Startups: 2 (NovaSaaS + MedQuick)
- Total Registered Users: 4
- Average Health Score: ~52 (average of 82 and 22)
- Platform Revenue: aggregated

Startup List:

- Both startups listed with health scores, risk badges, last activity date

User List:

- All 4 users: Arjun Mehta (founder), Priya Nair (founder), Rohit Sharma (investor), Sunita Das (admin)

3. Startup Detail — Click-Through (`/admin/startup/:id`)

1. Click on **NovaSaaS Technologies** row
2. Full detail page:
   - Startup Profile: company info, sector, stage, team size, CIN
   - Health Score + all 5 component breakdowns
   - All metrics (latest period shown)
   - All milestones list
3. Go back → click **MedQuick Health**
4. See critical health score, 2-month runway, high churn — critical risk

4. Change Subscription Plan

1. On the Admin Dashboard, find **Priya Nair** or **MedQuick Health**
2. Click the plan selector → change from **Free** → **Pro**
3. ✅ Plan updated — Priya can now access Forecasting, AI Advisor, Reports
4. Change NovaSaaS from **Enterprise** → **Pro** → change back to **Enterprise**

📝 Additional Edge Cases to Test

TEST	HOW TO TRIGGER
Account lockout	Enter wrong password 5× on any account
Auto-unlock	Wait 15 minutes OR run `npm run seed` in `/backend`
Cache invalidation	Submit new metrics → visit `/forecasting` → new AI call made
AI fallback mode	Remove OpenRouter key from `.env`, restart backend → Forecasting still works (rule-based)
Simulation extremes	Set Revenue Change to -50% → watch health score hit 0
Empty state	Register a brand new account → all pages show "no data" empty states
OKR slider	On any OKR's key result, drag the current value slider → auto-saves
Report PDF	Open Reports → click Print → choose "Save as PDF" in print dialog
CSV Export	Click Export CSV on Reports page → open the `.csv` in Excel
Alert dismiss	On Priya's dashboard, click ✕ on a critical alert → confirm it disappears

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🚀 Quick Start Checklist

```
1. Make sure servers are running:
   → Backend:  npm run dev  (in /backend)  → http://localhost:5000
   → Frontend: npm run dev  (in /frontend) → http://localhost:5173

2. Login order for full coverage:
   → founder1@nspms.in   (healthy founder — test all founder features)
   → founder2@nspms.in   (at-risk founder — test alerts, overdue milestones)
   → investor@nspms.in   (investor — test comparison feature)
   → admin@nspms.in      (admin — test startup management)

3. All passwords: password123
```